EXP.NO: 10	Develop an application to change the font and color of the text and display toast message when the user presses the button.
DATE:	

AIM:

To develop an Android application that allows the user to change the font and color of text displayed on the screen and display a Toast message when a button is pressed.

ALGORITHM:

1. **Initialize the Android Application:**
   - Set up a new Android project in **Android Studio**.
   - Create a **MainActivity** Java file for the main screen of the application.
2. **Design the User Interface (UI):**
   - In the **XML layout file**, add the following UI components:
     - **TextView**: For displaying the text whose font and color will be changed.
     - **Button**: To trigger the change in font and color, and show a **Toast message**.
     - **Spinner or Color Picker**: To let the user choose a color (optional, or a simple button for a predefined color change).
     - **RadioButtons** or a **Dropdown Menu** to select different font styles (optional).
3. **Declare Variables:**
   - Create references for the **TextView**, **Button**, **Color Picker** (if used), and **Radio Buttons** (if used to change font style).
   - Define variables for the **font style**, **color**, and the **message** to display in the **Toast**.
4. **Set Event Listener for the Button:**
   - Set an **OnClickListener** for the **Button**.

- Inside the onClick() method, write the logic to:
  - Change the **font** of the **TextView** based on the selected font style.
  - Change the **color** of the **TextView** based on the selected color.
  - Display a **Toast message** indicating the action has been performed.

5. **Implement Font Style Changes:**
   - Change the **TextView's font** using setTypeface() to apply a selected font style (e.g., bold, italic).
   - You can either use predefined styles (like Typeface.BOLD or Typeface.ITALIC) or let the user choose from a list of font styles.
6. **Implement Color Change:**
   - Change the **TextView's text color** by using the setTextColor() method and applying the selected color (e.g., Color.RED, Color.BLUE, or a color from the **Color Picker**).
   - If using a color picker or dropdown, get the selected color and set it on the **TextView**.
7. **Show Toast Message:**
   - Display a **Toast** message to the user when the button is clicked, informing them that the text style or color has changed. You can also show a message like "Font and color changed!" or any other appropriate message.
8. **End the Operation:**
   - After the changes are made, the application will display the updated **TextView** with the new font and color, and the **Toast message** will appear to provide feedback.

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"
```

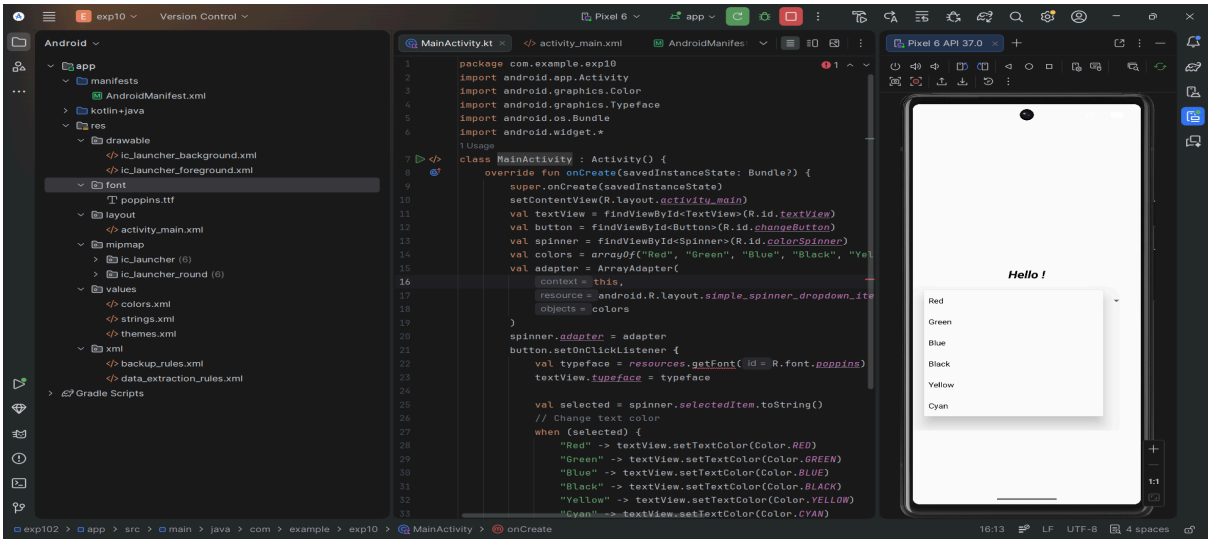
```
    android:gravity="center"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:id="@+id/textView"
        android:text="Hello Android!"
        android:textSize="24sp"
        android:layout_marginBottom="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
    <Spinner
        android:id="@+id/colorSpinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/changeButton"
        android:text="Change Style"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

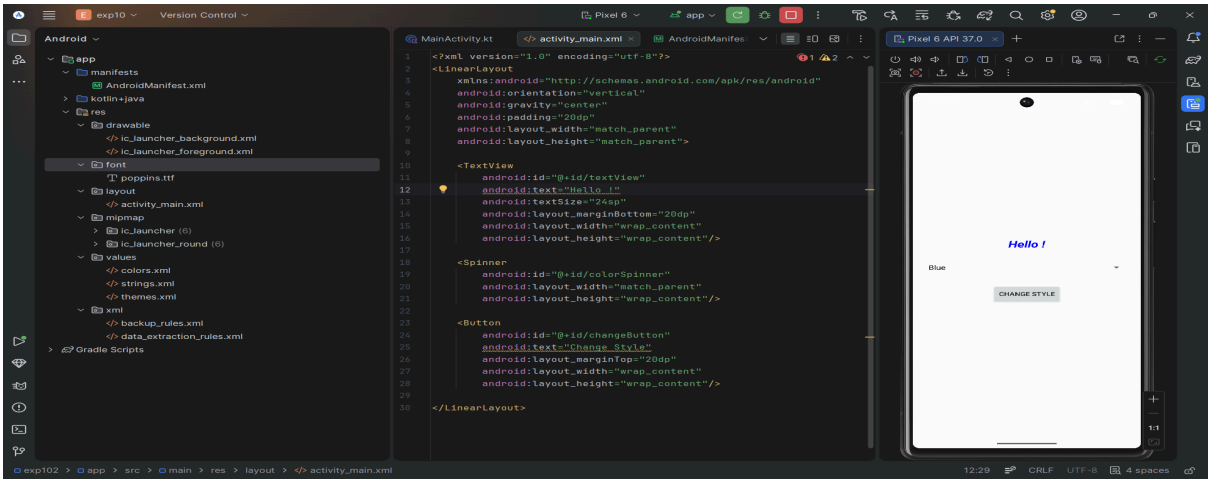
MainActivity.kt:

```
package com.example.exp10
import android.app.Activity
import android.graphics.Color
import android.graphics.Typeface
import android.os.Bundle
```

```
import android.widget.*
class MainActivity : Activity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        val textView = findViewById<TextView>(R.id.textView)
        val button = findViewById<Button>(R.id.changeButton)
        val spinner = findViewById<Spinner>(R.id.colorSpinner)
        val colors = arrayOf("Red", "Green", "Blue", "Black")
        val adapter = ArrayAdapter(
            this,
            android.R.layout.simple_spinner_dropdown_item,
            colors
        )
        spinner.adapter = adapter
        button.setOnClickListener {
            val selected = spinner.selectedItem.toString()
                when (selected) {
                "Red" -> textView.setTextColor(Color.RED)
                "Green" -> textView.setTextColor(Color.GREEN)
                "Blue" -> textView.setTextColor(Color.BLUE)
                "Black" -> textView.setTextColor(Color.BLACK)
            }
            textView.setTypeface(null, Typeface.BOLD_ITALIC)
            Toast.makeText(this, "Style Applied!", Toast.LENGTH_SHORT).show()
        }
    }
}
```

Output:





RESULT:

An mobile application is developed in Android studio to change the font and color of the text and display toast message when the user presses the button.